

Cholesterol, Total

Test Details

METHODOLOGY

Enzymatic

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

RELATED INFORMATION

- [Cardiovascular Disease Risk Assessment Tools \(/node/462\)](/node/462)
 - [Clinical Conditions Associated With Cardiovascular Disease](#)
([../220/documents/3542362](#))
 - [Clinical Utility Test List \(/node/463\)](/node/463)
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Use

Evaluate lipid status and metabolic disorders. High levels of cholesterol that reflect high levels of HPLs may be caused by an inherited defect in lipoprotein metabolism, by disease of the endocrine system, by liver disease, or by renal disease. Low levels of cholesterol in the plasma may reflect an inherited deficiency of either LDL or HDL, or they may reflect impairment of liver function. Various hormone conditions are also related to cholesterol levels. Increased serum cholesterol in hypothyroid persons shows an increased LDL and decreased HDL. Low cholesterols are found in cases of hyperthyroidism, severe liver disease, pernicious anemia, and with increased estrogens. Pregnancy is accompanied by a moderate increase. Cholesterol is increased in early hepatitis, obstructed bile ducts, primary biliary cirrhosis, nephrotic syndrome, and diabetic meningitis. Finally, through much controversy, it appears that cholesterol is implicated in atherosclerosis and heart disease. Evaluate risk of coronary arterial occlusion, atherosclerosis, myocardial infarction, and complications including the demise of the patient.

Increased in primary hypercholesterolemia, secondary hyperlipoproteinemias including nephrotic syndrome, hypothyroidism, primary biliary cirrhosis, and some cases of diabetes mellitus. **Low levels** have been found in cases of malnutrition, malabsorption, hyperthyroidism, myeloma, macroglobulinemia of Waldenström, polycythemia vera, myeloid metaplasia, myelofibrosis, chronic myelocytic leukemia, analphalipoproteinemia (Tangier disease), abetalipoproteinemia (Bassen-Kornzweig syndrome) (acanthocytosis), and in some individuals who subsequently present with carcinoma. Levy points out that the weak inverse relationship with cancer, mostly colon carcinoma, is limited to cholesterol levels <190 mg/dL and is limited to men.¹ Hypocholesterolemia may occur with sideroblastic anemia or in the thalassemias.

Cholesterol relates to coronary heart disease risk.² Since premature mortality from coronary arterial disease is rampant and since cholesterol levels are available as a test which can detect a modifiable risk factor, serum cholesterol remains a critical and genuinely newsworthy topic and an important test. Effective intervention is available when cholesterol studies identify subjects likely to benefit, asymptomatic persons as well as those with recognized coronary disease.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age (y)	Acceptable	Borderline	High
0 to 19	<170 (or 100–169)	170–199	≥200
>19	<200 (or 100–199)	200–239	≥240

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Improper labeling

References

Abell LL, Levy BB, Brodie BB, Kendall FE. A simplified method for the estimation of total cholesterol in serum and demonstration of its specificity. *J Biol Chem*. 1952 Mar; 195(1):357-366. PubMed 14938387

(<http://www.ncbi.nlm.nih.gov/pubmed/14938387>)

Footnotes

1. Levy RI. Cholesterol and disease—what are the facts? *JAMA*. 1982 Dec 3; 248(21):2888-2890. PubMed 7143656 (<http://www.ncbi.nlm.nih.gov/pubmed/7143656>)
2. Rifkind BM, Segal P. Lipid research clinics program reference values for hyperlipidemia and hypolipidemia. *JAMA*. 1983 Oct 14; 250(14):1869-1872. PubMed 6578354 (<http://www.ncbi.nlm.nih.gov/pubmed/6578354>)